

Laboratory Manual

Course Title (Course Code)

6th Semester Spring 2024



BS in Computer Science
Department of Computer Science & Information Technology
Sir Syed University of Engineering and Technology
University Road Karachi - 75300
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Course Learning Outcome(s):

CLO #	Course Learning Outcomes (CLO's)	Bloom's Taxonomy Level
1		
2		
3		

Laboratory Experiments

Lab #	Topics	Page #
1		
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Laboratory Rubrics

Criteria	Exceeds Expectations (≥90%)	Meets Expectations (70%-89%)	Developing (50%-69%)	Unsatisfactory (<50%)
Software Handling (2 Marks)	Able to use software with its standard and advanced features without assistance	Able to use software with its standard and advanced features with minimal assistance	Able to use software with its standard features with assistance	Unable to use the software
Programming/ Simulation (5 Marks)	Able to program/ simulate the lab tasks with simplification	Able to program/ simulate the lab tasks without errors	Able to program/ simulate lab tasks with errors	Unable to program/ simulate
Results (2 Marks)	Able to achieve all the desired results with alternate ways to improve measurements	Able to achieve all the desired results	Able to achieve most of the desired results with errors	Unable to achieve the desired results
Lab Report (1 Mark)	Laboratory manual has no grammatical and/ or spelling errors. All sections of the report are very well written and technically accurate.	Laboratory manual has very few grammatical/ spelling errors. All sections of the report are technically accurate.	-Laboratory manual has multiple grammatical/ spelling errors. Few sections of the report contain technical error s.	Laboratory manual has several grammatical/ spelling errors and sentence construction is poor. -All sections of the report contain multiple technical errors.

Laboratory Score

Lab#	Criteria				Score
1.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
2.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
3.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
4.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
5.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
6.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
7.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
8.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
9.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
10.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
11.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
12.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
13.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
14.	Software Handling ()/2	Programming/ Simulations ()/5	Results ()/2	Lab Report ()/1	
Obtained Score out of (140)					
Obtained Score out of (X) = Obtained Score out of (140) /140* X					

Open Ended Lab Rubrics

Criteria	Exceeds Expectations ($\geq 90\%$)	Meets Expectations (70%-89%)	Developing (50%-69%)	Unsatisfactory ($< 50\%$)	Score
Experiment Design & Planning	Demonstrates a thorough understanding of the problem and develops a well-defined objective, and detailed experimental procedure with logical justification.	Demonstrates a good understanding of the problem and develops a clear objective, and experimental procedure with adequate justification.	Demonstrates some understanding of the problem and develops an objective, and experimental procedure, but with limited justification.	Demonstrates limited understanding of the problem and develops a vague objective, and experimental procedure with weak justification.	
Execution and Data Collection	Executes the experiment effectively, following the designed procedure and accurate recording of observations, measurements, and parameters.	Executes the experiment well, following the designed procedure and accurate recording of most observations, measurements and parameters.	Executes the experiment adequately, following the designed procedure and recording of some observations, measurements, and parameters.	Executes the experiment poorly, deviating from the designed procedure and inaccurate recording of observations, measurements, and parameters.	
Analysis and Interpretation	Organizes and presents observations in a clear and concise manner, draws valid conclusions.	Organizes and presents observations effectively, draws reasonable conclusions.	Organizes and presents observations adequately, draws some conclusions.	Organizes and presents data poorly, draws weak conclusions.	
Communication and Presentation	Presents the experiment and its findings in a well-structured manner, uses precise scientific terminology and language, and provides clear explanations of concepts, procedures, and results.	Presents the experiment and its findings effectively, use appropriate scientific terminology and language, and provide clear explanations of most concepts, procedures, and results.	Presents the experiment and its findings adequately, use basic scientific terminology and language, and provide satisfactory explanations of some concepts, procedures, and results.	Presents the experiment and its findings poorly, lacks organization and clarity, uses inappropriate terminology and language, and Provides weak explanations of concepts, procedures, and results.	

Semester Project Rubric

Criteria	Exceeds Expectations (>=90%)	Meets Expectations (70%-89%)	Developing (50%-69%)	Unsatisfactory (<50%)	Score
Project Demonstration (4 Marks)	☐ Able to demonstrate the project with achievement of required objectives having clear understanding of project limitations and future enhancements. Hardware and/or Software modules are fully functional, if applicable.	☐ Able to demonstrate the project with achievement of required objectives but understanding of project limitations and future enhancements is insufficient. Hardware and/or Software modules are functional, if applicable.	☐ Able to demonstrate the project with achievement of at least 50% required objectives and insufficient understanding of project limitations and future enhancements. Hardware and/or Software modules are partially functional, if applicable.	☐ Able to demonstrate the project with achievement of less than 50% required objectives and lacks in understanding of project limitations and future enhancements. Hardware and/or Software modules are not functional, if applicable.	
Project results (3 Marks)	☐ Able to achieve all the desired results with alternate ways to improve measurements.	☐ Able to achieve all the desired results.	☐ Able to achieve most of the desired results with errors.	☐ Unable to achieve the desired results.	
Report Writing (3 Marks)	☐ Project report has no grammatical and/ or spelling errors. All sections of the report are very well-written and technically accurate.	☐ Project report has very few grammatical/ spelling errors. All sections of the report are technically accurate.	☐ Project report has multiple grammatical/ spelling errors. Few sections of the report contain technical errors.	☐ Project report has several grammatical/ spelling errors and sentence construction is poor.	
Viva (5 Marks)	☐ Able to answer the questions easily and correctly across the project.	☐ Able to answer the questions related to the project.	☐ Able to answer the questions but with mistakes.	☐ Unable to answer the questions.	
Obtained Score out of (15)					
Obtained Score out of (X)= Obtained Score out of (15) /15* X					

Affective Domain Rubrics

Criteria	Excellent (100% - 85%)	Good (84% - 75%)	Fair (74% - 60%)	Poor (Less than 60%)
Speaks Clearly	Speaks clearly and distinctly all the time, and confidently.	Speaks clearly and distinctly most of the time, but is confused for a brief period of time, however, recovers.	Speaks clearly and distinctly most of the time, but seems not confident about what has been delivered. Shows lack of confidence.	Often mumbles or cannot be understood and clearly lacks confidence in delivering the content
Points:				
Preparedness	Student is completely prepared and has obviously rehearsed.	Student seems pretty prepared but might have needed a couple more rehearsals.	The student is somewhat prepared, but it is clear that rehearsal was lacking.	Student does not seem at all prepared to present.
Points:				
Answer back	Student calmly listens to the questions and responds to the question confidently and correctly	Student calmly listens to the questions, responds confidently but some of the responses are incorrect.	Student shows anxiety while listening to the questions, and gives some correct responses, but some of the responses are incorrect.	Student shows anxiety while listening to the questions, and most of the responses are incorrect.
Points:				
Posture, Eye Contact & Speaking Volume	Stands up straight, looks relaxed and confident. Establishes eye contact with everyone in the room during the presentation. Volume is loud enough to be heard by all members in the audience throughout the presentation.	Stands up straight and establishes eye contact with everyone in the room during the presentation. Volume is loud enough to be heard by the audience, but is sometimes not audible.	Sometimes stands up straight and establishes eye contact. Volume is loud enough to be heard by the audience, but many sentences spoken are not clear.	Lazy and informal posture. Does not look at people during the presentation. Volume is also too soft to be heard by the audience.
Points:				
Attitude	Never is publicly critical of the project or the work of others. Always has a positive attitude about the tasks.	Rarely is publicly critical of the project or the work of others. Often has a positive attitude about the tasks.	Occasionally is publicly critical of the project or the work of other members of the group. Usually has a positive attitude about the tasks.	Often is publicly critical of the project or the work of other members of the group. Often has a negative attitude about the tasks).
Points:				

Laboratory Examination Rubric

Criteria	Exceeds Expectations ($\geq 90\%$)	Meets Expectations (70%-89%)	Developing (50%-69%)	Unsatisfactory ($< 50\%$)	Score
Performance (15 Marks)	☐ Able to present full knowledge of both problem and solution.	☐ Able to present adequate knowledge of both problem and solution	☐ Able to present sufficient knowledge of both problem and solution	☐ No or very less knowledge of both problem and solution	
Viva (5 Marks)	☐ Able to answer the questions easily and correctly across the project.	☐ Able to answer the questions related to the project	☐ Able to answer the questions but with mistakes	☐ Unable to answer the questions	
Obtained Score out of (20)					
Obtained Score out of (X)= Obtained Score out of (20) /20* X					

Affective Domain Rubrics

Criteria	Excellent (100% - 85%)	Good (84% - 75%)	Fair (74% - 60%)	Poor (Less than 60%)
Speaks Clearly	Speaks clearly and distinctly all the time, and confidently.	Speaks clearly and distinctly most of the time, but is confused for a brief period of time, however, recovers.	Speaks clearly and distinctly most of the time, but seems not confident about what has been delivered. Shows lack of confidence.	Often mumbles or cannot be understood and clearly lacks confidence in delivering the content
Points:				
Preparedness	Student is completely prepared and has obviously rehearsed.	Student seems pretty prepared but might have needed a couple more rehearsals.	The student is somewhat prepared, but it is clear that rehearsal was lacking.	Student does not seem at all prepared to present.
Points:				
Answer back	Student calmly listens to the questions and responds to the question confidently and correctly	Student calmly listens to the questions, responds confidently but some of the responses are incorrect.	Student shows anxiety while listening to the questions, and gives some correct responses, but some of the responses are incorrect.	Student shows anxiety while listening to the questions, and most of the responses are incorrect.
Points:				
Posture, Eye Contact & Speaking Volume	Stands up straight, looks relaxed and confident. Establishes eye contact with everyone in the room during the presentation. Volume is loud enough to be heard by all members in the audience throughout the presentation.	Stands up straight and establishes eye contact with everyone in the room during the presentation. Volume is loud enough to be heard by the audience, but is sometimes not audible.	Sometimes stands up straight and establishes eye contact. Volume is loud enough to be heard by the audience, but many sentences spoken are not clear.	Lazy and informal posture. Does not look at people during the presentation. Volume is also too soft to be heard by the audience.
Points:				
Attitude	Never is publicly critical of the project or the work of others. Always has a positive attitude.	Rarely is publicly critical of the project or the work of others. Often has a positive attitude.	Occasionally is publicly critical of the project or the work of other members of the group. Usually has a positive attitude.	Often is publicly critical of the project or the work of other members of the group. Often has a negative attitude.
Points:				

Final Lab Assessment

Criteria	Score
Laboratory (15)	
Semester Project (15)	
Laboratory Examination (20)	
Total Score out of (50)	

Examiner: _____

(Name and Signature)